

Ryder Fine

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Design Engineer — Product Design & Additive Manufacturing

EDUCATION

COLORADO SCHOOL OF MINES

Golden, CO

B.S. Design Engineering

May 2027

Relevant Coursework: Design for Additive, Mechanics of Composite Materials, Prototyping for Concept Validation

Club Water Polo; Dean's List 2024–2026; Oredigger Leadership Community

UNIVERSITY OF NICOSIA

Nicosia, Cyprus

Study Abroad — Fluids, Circuits, Manufacturing; international project teams

Jan 2025 – Jun 2025

EXPERIENCE

EMERSON ELECTRIC (DP FLOW)

Boulder, CO

Manufacturing Engineering Intern

May 2026 – Aug 2026

- Evaluated laser engraving software feasibility and cost, then pivoted to an automated alternative saving \$5,000/yr
- Designed workholding fixtures in PTC Creo for prototype product line, applying GD&T to critical features
- Authored process documentation and work instructions for 5 production operations (welding, machining, and assembly)

LABRIOLA INNOVATION HUB

Golden, CO

Technical Lead

Jan 2024 – Present

- Selected and set up \$60,000 of engineering equipment (CNC, additive, composites) for cross-functional teams
- Developed user training system, standard operating procedures, and website serving 7,000 users
- Built and deployed an inventory management system tracking 40 SKUs, used by 20 staff to replace manual spreadsheets

INDEPENDENT 3D PRINTING BUSINESS

Trophy Club, TX

Founder

Jun 2020 – Aug 2024

- Built and managed an online marketplace selling 3D printed replacement parts, completing 100+ sales
- Upgraded and maintained an Ender 3 and rebuilt broken CR-10 printers

PROJECTS & RESEARCH

ROBOTIC SPACE EXPLORATION (RoSE) LAB

Golden, CO

Undergraduate Researcher, Lunar Rover Wheel Testing Platform

Feb 2026 – Present

- Designed storage/lift subsystem stowing a 550 lb rail-and-gantry assembly overhead on dual half-ton chain hoists, cutting stow time by 40% in shared mixed-use space
- Verified factor of safety (FoS) > 5 through chain load and beam deflection analysis
- Developed system concept of operations (ConOps) and a \$780 bill of materials
- Implemented clamps and safety net to eliminate the top-severity subsystem risk

ALTERNATIVE MINING PRACTICES SYSTEMS STUDY

Golden, CO

Course Project, Systems Modeling and Design (EDNS 310)

Aug 2025 – Dec 2025

- Built quantitative system dynamics models in Vensim to test feedback loops and long-term resource impacts
- Evaluated alternative material sourcing methods and recommended automated mining equipment to decrease environmental impact and safety incidents

SKILLS

ENGINEERING METHODS: FEA, DFM/DFA, technical drawings, rapid prototyping and testing, value stream mapping, topology optimization, technical writing, GD&T (ASME Y14.5), tolerance stack-up analysis

MANUFACTURING: CNC machining, welding, additive mfg, casting (sand, investment, die), machine shop, wood shop

SOFTWARE: SolidWorks, SolidWorks Simulation (FEA), PTC Creo, Onshape, MATLAB, Python, MS Office Suite, Claude Code, SysML (CAMEO), LabVIEW

CERTIFICATIONS & ACCOLADES

Certified SolidWorks Associate (CSWA), Apr 2024 | **Markforged Certified Additive Expert — Composites**, Jul 2024

Eagle Scout, Boy Scouts of America, Jun 2023 | **Hazardous Waste Generator**, Mines EHS, Aug 2026